BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - II - May - 2026

II Semester

22MA201 & ENGINEERING MATHEMATICS II

1	(i)	A robot follows path r(t) = ti + t^2j Determine velocity vector and interpret motion. (3 Marks - [U/C, 2])
2	(ii)	Differentiate scalar and vector line integrals. (2 Marks - [U/C, 1])
3	(i)	In a heat flow system, $\vec{F} = (x+y)\hat{i} + (x-y)\hat{j}$. Analyze the integrand in Green's theorem. (3 Marks - [U/C, 2])
4	(ii)	In a circulation model, what does a positive value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ indicate? (2 Marks - [U/C, 1])
5	(i)	A force field acts on a particle moving along a curve. Analyze and interpret the physical meaning of the line integral. (3 Marks - [U/C, 2])
6	(ii)	Define path independence in engineering systems. (2 Marks - [U/C, 1])
7	(i)	In a fluid flow model, $\vec{F} = y\hat{i} + x\hat{j}$. Use Green's theorem to compute $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ and interpret its meaning. (3 Marks - [U/C, 2])
8	(ii)	In a fluid flow system, $\vec{F} = (x+y)\hat{i} + (2x-y)\hat{j}$. Compute $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$∂x∂Q−∂y∂P. (2 Marks - [U/C, 1])
9	(i)	Evaluate $\iint_S \vec{F}\cdot\hat{n}\, ds$ for the heat flux field $\vec{F} = x^2\,\hat{i} + yz\,\hat{j} + z^2\,\hat{k}$, where $\hat{n} = \hat{j}$ and the surface is $y = 1$, $0 \le x \le 2$, $0 \le z \le 1$. (5 Marks - [Ap/P, 3])
10	(i)	In an electromagnetic field, the vector field is $\vec{F} = z\hat{i} + x\hat{j} + y\hat{k}$. Determine the circulation around the boundary of the region in the yz-plane (x = 0) bounded by $0 \le y \le 1$, $0 \le z \le 1$ using Stokes' theorem. (5 Marks - [Ap/P, 3])

11	(i)	Evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$ for the fluid velocity field $\vec{F} = 2xy\,\hat{i} + yz\,\hat{j} + xz\,\hat{k}$, where $\hat{n} = \hat{i}$ and the surface is $x = 2$, $0 \le y \le 1$, $0 \le z \le 2$. (5 Marks - [Ap/P, 3])
12	(i)	In a heat transfer system, the flux field is $\vec{F} = xy\hat{i} + yz\hat{j} + zx\hat{k}$. Compute the circulation of heat flow around the boundary of the region in the xz-plane (y = 0) bounded by $0 \le x \le 1$, $0 \le z \le 1$ using Stokes' theorem. (5 Marks - [Ap/P, 3])
13	(i)	In an electrical field, $\vec{F} = (y)\hat{i} + (x)\hat{j}$. Use a line integral to determine the work done along the boundary of the square defined by $0 \le x \le 1$, $0 \le y \le 1$. (5 Marks - [Ap/P, 3])
14	(i)	Evaluate and interpret the total outward heat flux using divergence theorem for $F = (x^2 - yz)i + (y^2 - xz)j + (z^2 - xy)k$ over a rectangular domain $0 \le x \le a$, $0 \le y \le b$, $0 \le z \le c$. (5 Marks - [Ap/C, 3])
15	(i)	In a fluid transport system, the velocity field is $\vec{F} = (2x)\hat{i} + (y)\hat{j} + (z)\hat{k}$. Use a surface integral to compute the total flux through the surface $x = 1$ bounded by $0 \le y \le 2$, $0 \le z \le 1$. (5 Marks - [Ap/P, 3])
16	(i)	Evaluate and interpret the total outward heat flux using divergence theorem for $F = 4xz\,i - y^2\,j + yz\,k$ over a unit cubic control volume ($0 \le x,y,z \le 1$) (5 Marks - [Ap/P, 3])
17	(i)	A control response is modeled by $f(z) = z^2$. Analyze whether the function is differentiable at $z = 1 + i$. (3 Marks - [U/C, 2])
18	(ii)	A signal transformation is defined by $w = z^2$. Identify whether the function is single-valued or multi-valued and justify. (2 Marks - [U/C, 1])
19	(i)	In a fluid transport system, the stream function is given by $v(x,y)=2xy$. Determine the harmonic conjugate using the Milne–Thomson method. (3 Marks - [U/C, 2])
20	(ii)	In an electrical field application, the real part of an analytic function is $u(x,y) = x^2 + 2xy$. Find $f'(z)$ using Milne–Thomson method. Hint: $f'(z) = u_x(z,0) - iu_y(z,0)$ (2 Marks - [U/P, 2])
21	(i)	A communication signal is represented by a complex function f(z). Evaluate the relationship between differentiability and continuity in complex systems. (3 Marks - [U/C, 2])

22	(ii)	An impedance transformation is represented by $w = \sqrt{z}$. Analyze whether the function is single-valued or multi-valued. (2 Marks - [U/C, 1])
23	(i)	An electric field potential is represented by $u(x,y)=e^x \cos y$. Construct the analytic function using the Milne–Thomson method. (3 Marks - [U/C, 2])
24	(ii)	In a fluid flow model, the imaginary part of an analytic function is $v(x, y) = x^2 - y^2$. Using Milne–Thomson method, find $f'(z)$. Hint: $f'(z) = v_y(z, 0) + iv_x(z, 0)$ (2 Marks - [U/P, 2])
25	(i)	A communication signal is modeled by $f(z) = e^z$. Analyze whether the function is analytic and interpret its significance in signal amplification. (5 Marks - [Ap/C, 3])
26	(i)	A signal transformation system uses the mapping $w = (1+i)z$. Determine the image of the region $y > 2$ in the w-plane and analyze the transformation characteristics (5 Marks - [An/C, 3])
27	(i)	In a thermal distribution model, the temperature potential is represented by $f(z) = (x^2 - y^2 + 2x) + i(2xy + 2y)$. Verify whether the function is analytic using the Cauchy–Riemann equations and interpret the physical meaning. (5 Marks - [Ap/C, 3])
28	(i)	An electrostatic field boundary represented by $\|z\| = 2$ is transformed under $w = z + 1 + 4i$. Determine the image of the boundary and interpret the transformation. (5 Marks - [An/C, 3])
29	(i)	In an electrical field system, the imaginary part is $v(x, y) = 2xy$. Construct the analytic function $f(z)$. (5 Marks - [Ap/P, 3])
30	(i)	A control transformation maps the points $z_1 = 1, z_2 = i, z_3 = -1$ onto $w_1 = 0, w_2 = 1, w_3 = -1$. Determine the bilinear transformation. (5 Marks - [An/C, 3])

31	(i)	In a heat transfer application, determine whether $u(x, y) = x^2 - y^2$ is harmonic and construct its harmonic conjugate. (5 Marks - [Ap/P, 3])
32	(i)	A signal transformation system maps the points $z_1 = 0$, $z_2 = i$, $z_3 = -i$ onto $w_1 = 1$, $w_2 = -1$, $w_3 = 0$. Determine the bilinear transformation and interpret the engineering significance. (5 Marks - [An/C, 3])